

CS & IT ENGINEERING



Operating System

Memory Management

Lecture -7

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Recap of Previous Lecture



Doubt ?

Topic

Multilevel Paging



Topics to be Covered



Topic

Multilevel Paging

Topic

Frame Allocation



Topic : Question

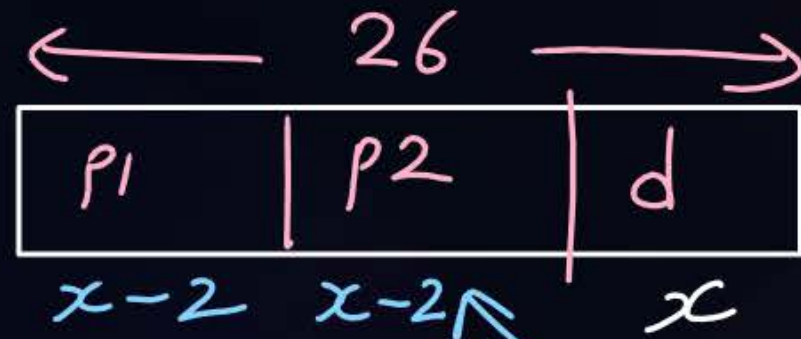
VA = 26 bits

Page table entry size = 4 bytes

2-level paging

Outer page table fits into a page exactly

Page size = 1 Kbytes?



$$2(x-2) + x = 26$$

$$2x - 4 + x = 26 \Rightarrow x = 10 \text{ bits}$$

$$\text{Page size} = 2^x \text{ B} \Rightarrow d = x \text{ bits}$$

no. of P.T. entries per page

$$= \frac{2^x \cancel{\text{B}}}{4 \cancel{\text{B}}}$$

$$= \frac{2^x}{2^2}$$

$$= 2^{x-2}$$

$$\Rightarrow \text{Page size} = 2^{10} \text{ B} = 1 \text{ KB}$$



Topic : Question



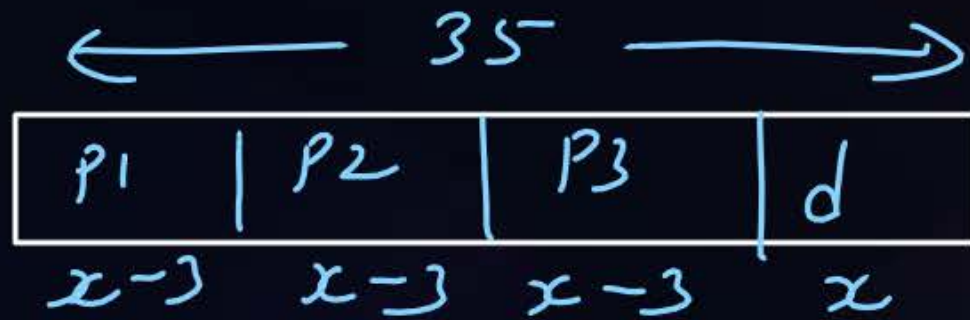
VA = 35 bits

Page table entry size = 8 bytes

3-level paging

Outer page table fits into a page exactly

Page size = 2 Kbytes?



$$\begin{aligned} 3(x-3) + x &= 35 \\ 4x &= 44 \Rightarrow x = 11 \Rightarrow \text{Page size} = 2^{11} \text{ B} = 2 \text{ KB} \end{aligned}$$



Topic : Question

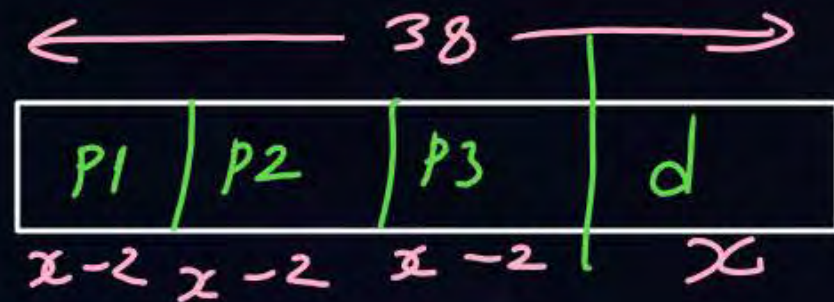
VA = 38 bits

Page table entry size = 4 bytes

3-level paging

Outer page table fits into a page exactly

Page size = 2 Kbytes?



$$3(x-2) + x = 38$$

$$4x = 44$$

$$x = 11$$

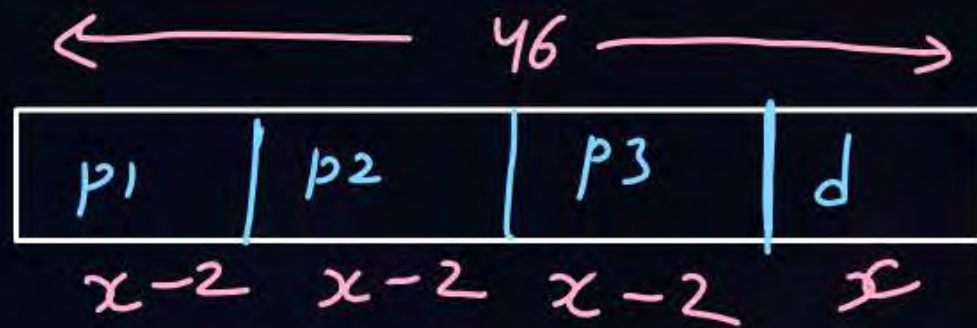
$$\Rightarrow \text{page size} = 2^{11} \text{ B} = 2 \text{ KB}$$



Topic : Question

#Q. A computer uses 46-bit virtual address, 32-bit physical address, and a three-level paged page table organization. The page table base register stores the base address of the first-level table (T1), which occupies exactly one page. Each entry of T1 stores the base address of a page of the second-level table (T2). Each entry of T2 stores the base address of a page of the third-level table (T3). Each entry of T3 stores a page table entry (PTE). The PTE is 32 bits in size. What is the size of a page in KB in this computer?

↓
4B



$$3(x-2) + x = 46$$

$$4x = 52$$

$$x = 13$$

$$\text{page size} = 2^{13} \text{ B} = 8 \text{ KB}$$

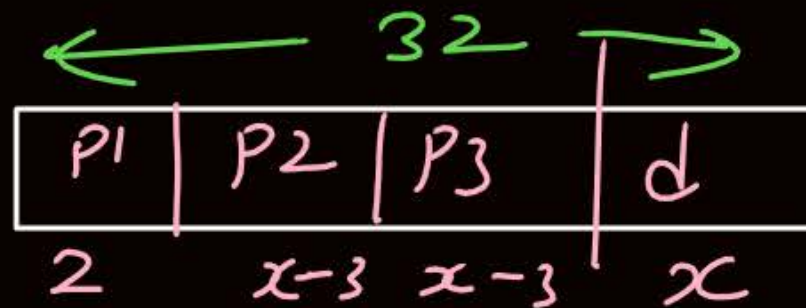
Ques) $VA = 32 \text{ bits}$

$P.T.E. = 8B$

3 levels Paging

outer PT stores only 4 P.T. entries

Page size = 4 KB



$$2 + 2(x-3) + x = 32$$

$$3x - 6 = 30$$

$$3x = 36$$

$$x = 12 \Rightarrow \text{Page size} = 4KB$$



$P1 = 2 \text{ bits}$



Topic : Question

Ans = 3KB

Size of page tables across all levels?

P1	P2	D
1	8	10

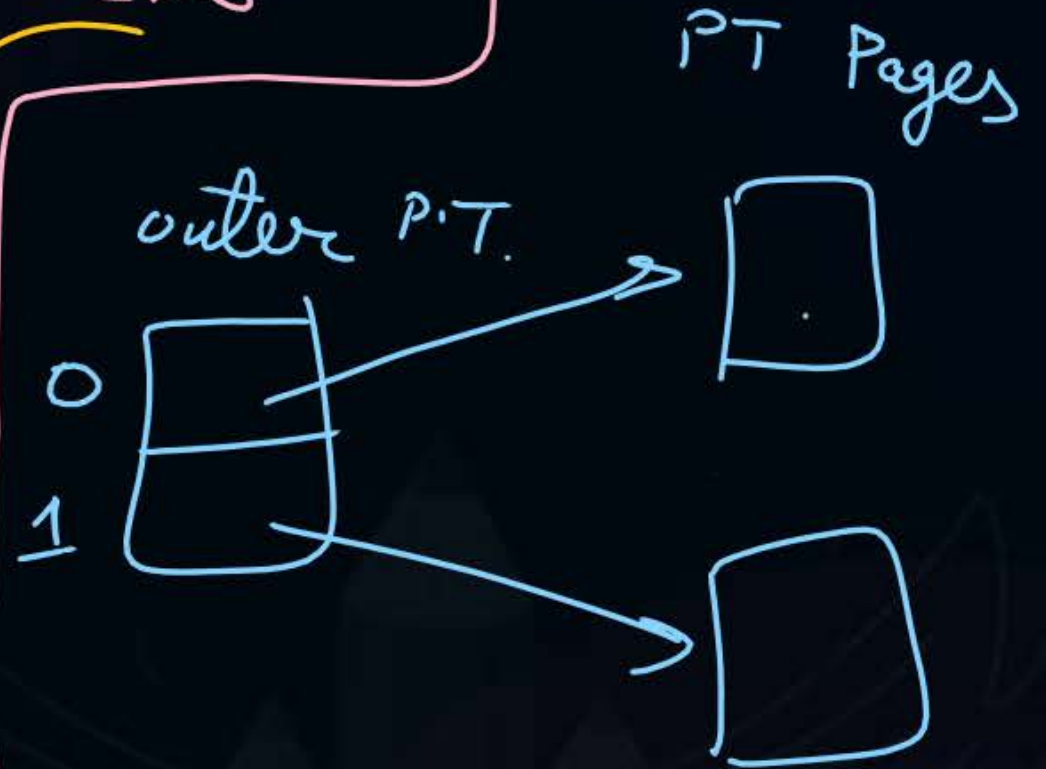
Page Size = $2^{10} = 1\text{KB}$

No. of pages to store outer P.T. = 1

Inner P.T. = $2^{P_1} = 2^1 = 2$

Total = 3

Total P.T. size
across all levels = $3 * 1\text{KB}$
= 3KB





Topic : Question

20 KB Ans

Size of page tables across all levels?

P1	P2	D
2	9	12

Page size = $2^{12} \text{ B} = 4 \text{ KB}$

no. of pages to store
outer P.T. = 1
inner PT = $2^2 = 4$
————— 11 —————
Total = 5 pages

Total PT size = $5 * 4 \text{ KB}$
 $= 20 \text{ KB}$



Topic : Question

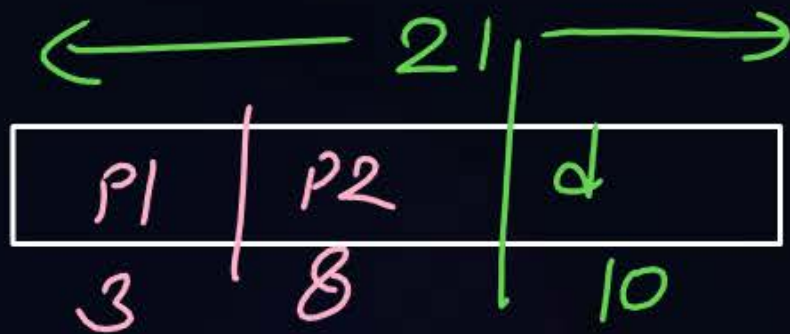
VA = 21 bits

Page table entry size = 4 bytes

2-level paging

Page size = 1 Kbytes.

Size of page tables across all levels?



$$\begin{array}{l} \text{no of pages to store outer PT} = 1 \\ \text{Inner PT} = 2^3 = 8 \\ \hline \text{Total} = 9 \end{array}$$

$$\begin{array}{l} \text{Total P.T. size} = 9 * 1 \text{ KB} \\ \text{across all levels} \\ = \underline{\underline{9 \text{ KB}}} \text{ Ans.} \end{array}$$



Topic : Question

Size of page tables across all levels?

P1	P2	P3	D
1	8	8	10

Page Size = 1KB



no. of pages to store outer P.T. = 1

middle PT = $2^{P_1} = 2^1 = 2$

inner PT = $2^{P_1} * 2^{P_2} = 2^1 * 2^8 = 2^9 = 512$

Total = 515

P.T. size = $515 * 1KB = \underline{\underline{515KB}}$ Ans.



Topic : Question

Size of page tables across all levels?

P1	P2	P3	D
2	7	7	10

Page size = $2^{10} \text{ B} = 1 \text{ KB}$

No. of pages for outer P.T. = 1

|| middle P.T. = $2^2 = 4$

|| inner P.T. = $2^2 * 2^7 = 512$

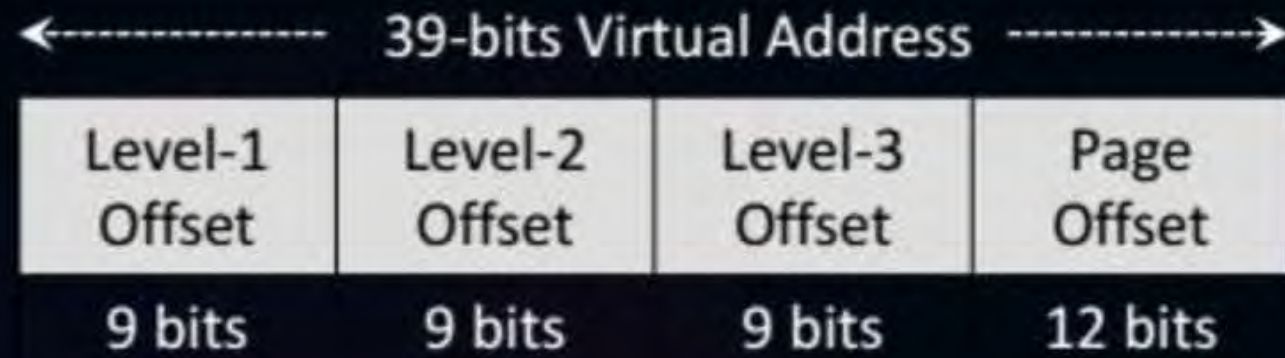
Total = 517 $\Rightarrow$ P.T. size = $517 * 1 \text{ KB}$
 $= 517 \text{ KB}$



Topic : Question

Ans = 4108

#Q. Consider a three-level page table to translate a 39-bit virtual address to a physical address as shown below:



The page size is 4 KB = (1KB = 2^{10} bytes) and page table entry size at every level is 8 bytes. A process P is currently using 2^{31} B $\Rightarrow$ LA = 31 bits (1 GB = 2^{30} bytes) virtual memory which OS mapped to 2 GB of physical memory. The minimum amount of memory required for the page table of P across all levels is _____ KB across all levels?



Topic : Solution



31

P_1	P_2	P_3	d
1	9	9	12

Pages for outer PT = 1

—||— middle PT = $2^1 = 2$

—||— Inner PT = $2^1 \cdot 2^9 = 1024$

Total = 1027

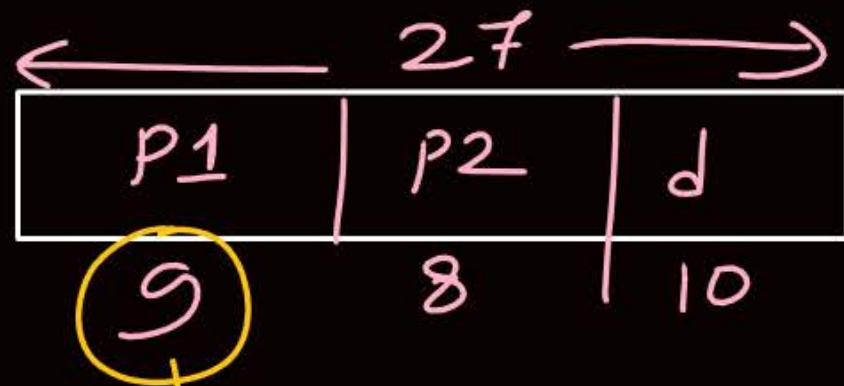
Total P.T. size = $1027 * 4KB$
= 4108 KB

Ques) 2-level paging:

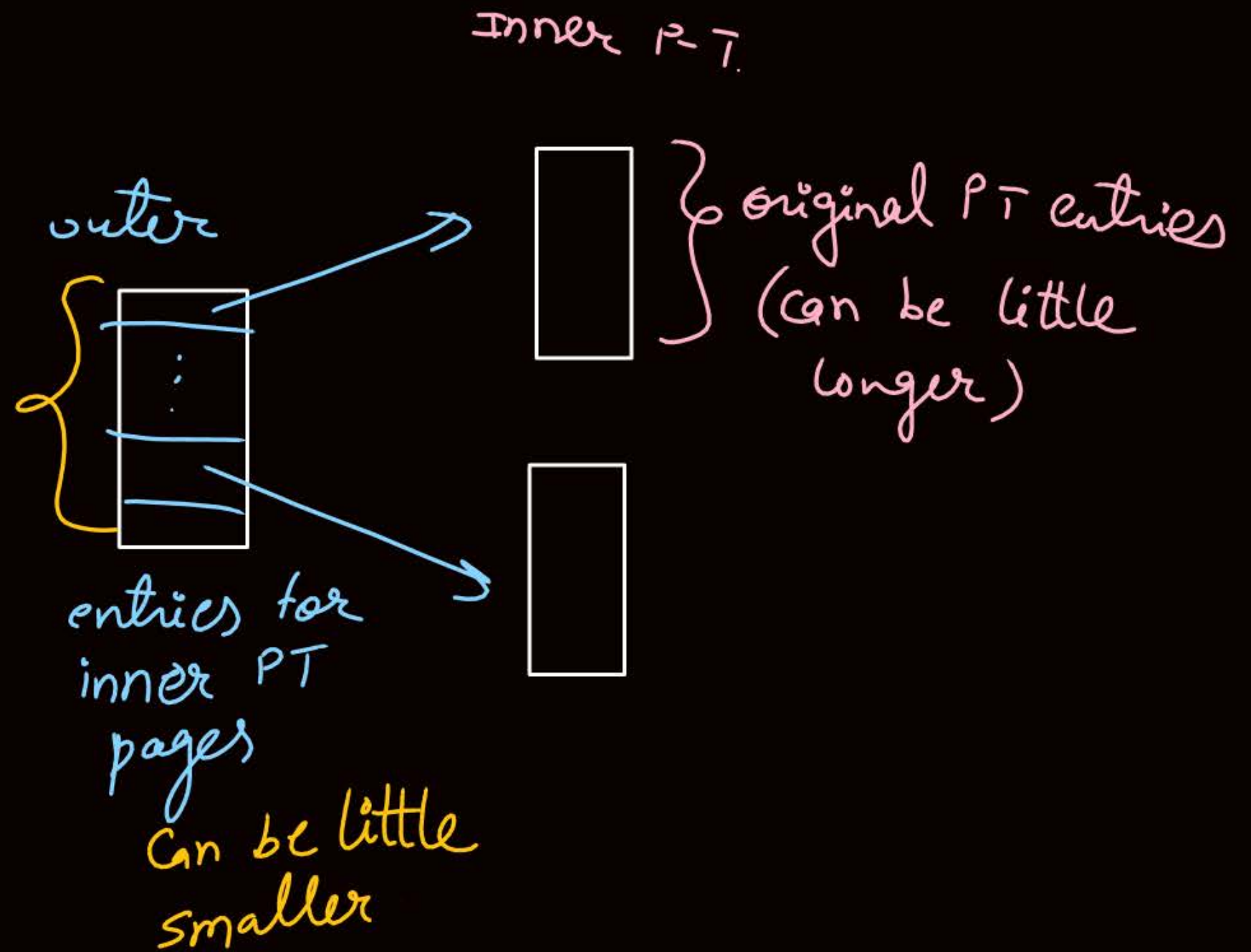
$LA = 27$ bits

Page size = 1 KB

P.T.E. = 4B



only when levels = 2
already fixed





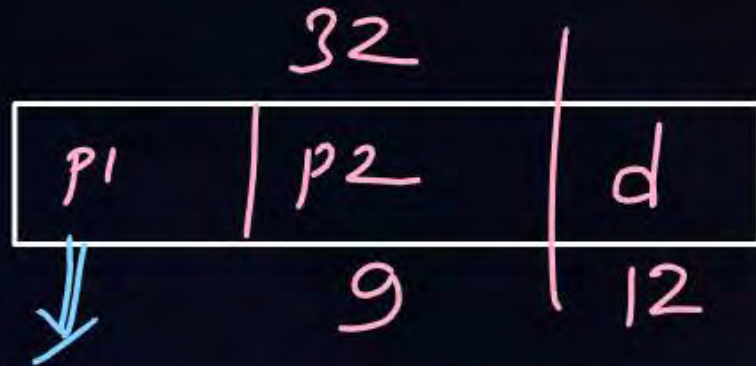
Topic : Question

[GATE-2025]



$$2^{12} \quad LA = 32 \text{ bits}$$

- #Q. A computer system supports a logical address space of 2^{32} bytes. It uses two-level hierarchical paging with a page size of 4096 bytes. A logical address is divided into a b-bit index to the outer page table, an offset within the page of the inner page table, and an offset within the desired page. Each entry of the inner page table uses eight bytes. All the pages in the system have the same size. The value of b is 11?



$$32 - (9 + 12) = 11$$

why multilevel paging:-

single level paging $\Rightarrow$ entire PT should be in mm $\Rightarrow$ It takes lot of space

multilevel paging $\Rightarrow$ only required pages of PT $\Rightarrow$ It saves lot of space in mm
are kept in mm

$\Downarrow$
using demand paging
for P.T. pages



#Q. Consider a 32-bit system with 4 KB page size and page table entries of size 4 bytes each. Assume $1 \text{ KB} = 2^{10}$ bytes. The OS uses a 2-level page table for memory management, with the page table containing an outer page directory and an inner page table. The OS allocates a page for the outer page directory upon process creation. The OS uses demand paging when allocating memory for the inner page table, i.e., a page of the inner page table is allocated only if it contains at least one valid page table entry.

An active process in this system accesses 2000 unique pages during its execution, and none of the pages are swapped out to disk. After it completes the page accesses, let X denote the minimum and Y denote the maximum number of pages across the two levels of the page table of the process.

The value of $X + Y$ is _____?

EMAT for multilevel paging:-

It increases due to PT access for multiple times.

for L level Paging

$$P.T. \text{ access time} = L * t_{mm}$$



Topic : Frame Allocation

- How many frames do we allocate per process?





Topic : Frame Allocation



How many frames do we allocate per process?

- If it is a single-user, single-tasking system, it's simple – all the frames belong to the user's process





Topic : Frame Allocation

2 Questions

- What is the minimum number of frames that a process needs?
- Is page replacement global or local?





Topic : Frame Allocation

1. Equal Allocation $\Rightarrow$ allocate equal no. of frames to each process
2. Proportional Allocation $\Rightarrow$ allocate frames as per size of process

ex:-

2 Processes
 $P_1 \Rightarrow 8$ pages
 $P_2 \Rightarrow 4$ pages
no. of frames = 6

Equal allocation

$$P_1 = \frac{6}{2} = 3 \text{ frames}$$

$$P_2 = \frac{6}{2} = 3 \text{ frames}$$

proportional allocation

$$P_1 = \frac{8}{12} * 6 = 4$$

$$P_2 = \frac{4}{12} * 6 = 2$$



Topic : Minimum Number of Frames



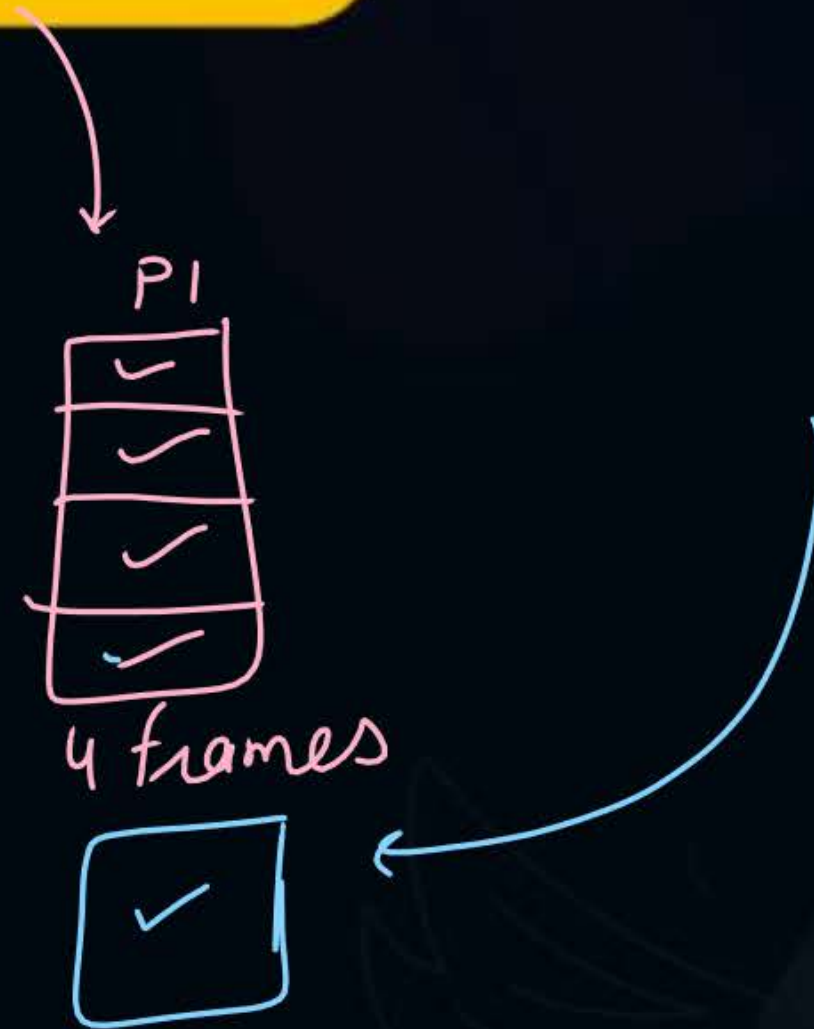
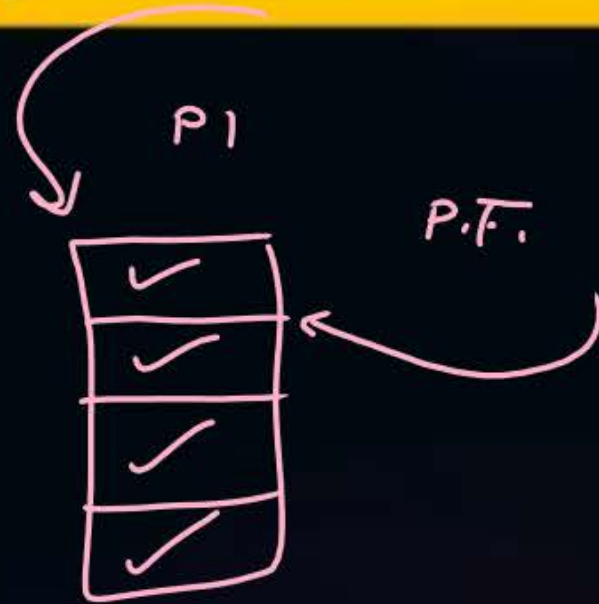
Every process must have enough pages to complete an instruction.

$\text{min size of page} \geq \text{size of instruction}$





Topic : Local Vs Global Allocation





Topic : Local Allocation

1. Local replacement requires that the page being replaced be in a frame belonging to the same process
2. The number of frames belonging to the process will not change
3. This allows processes to control their own page fault rate

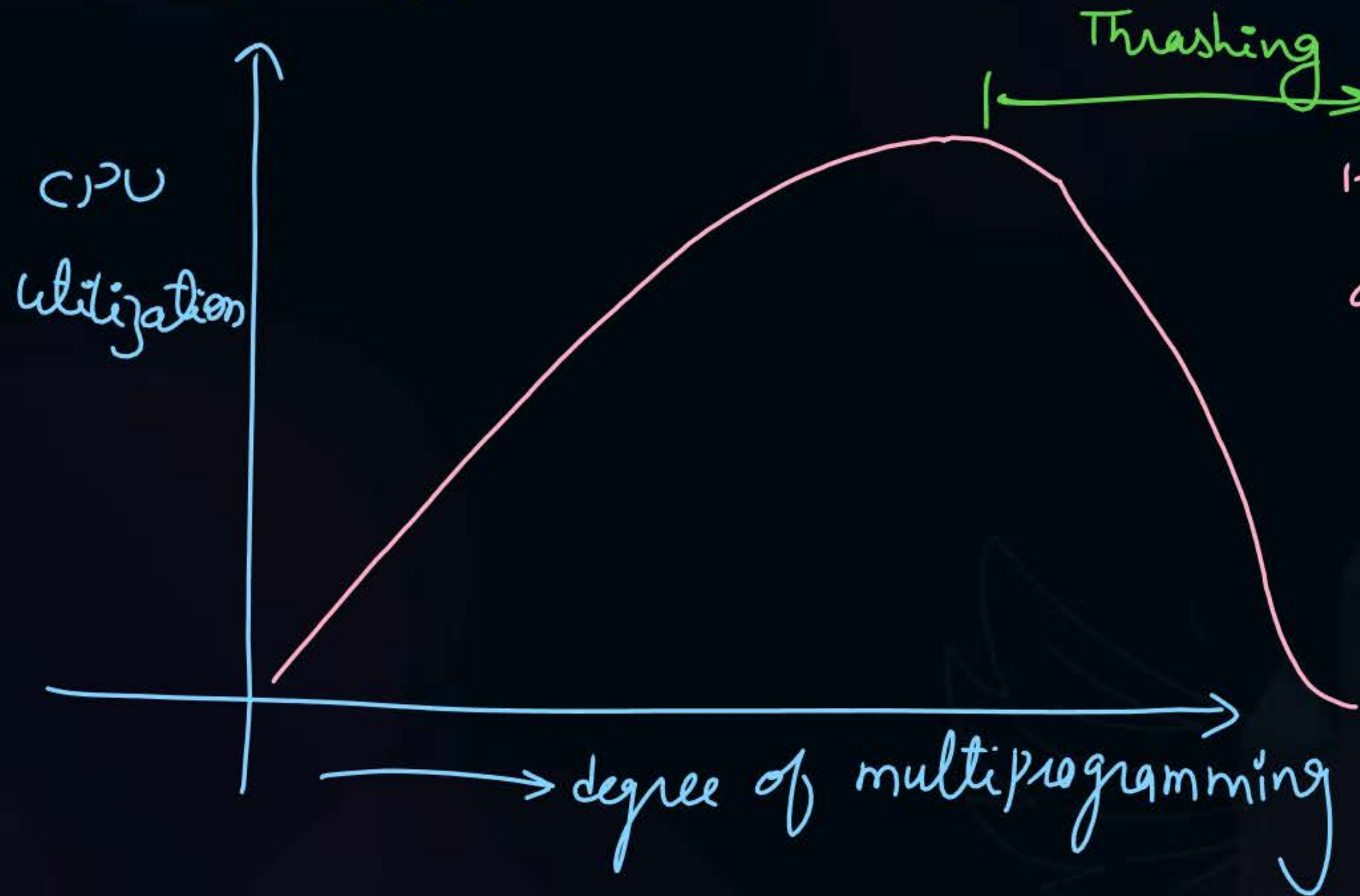


Topic : Global Allocation

1. The process can replace a page from a set that includes all the frames allocated to user processes
2. High-priority processes can increase their allocation at the expense of lower-priority processes
3. Global allocation makes for more efficient use of frames and their better throughput



Topic : Thrashing



high level paging activity (CPU spends more time for p.f. service than process execution).



Topic : Locality Model

1. Working Set Model
2. Page Fault Frequency



Topic : Working Set

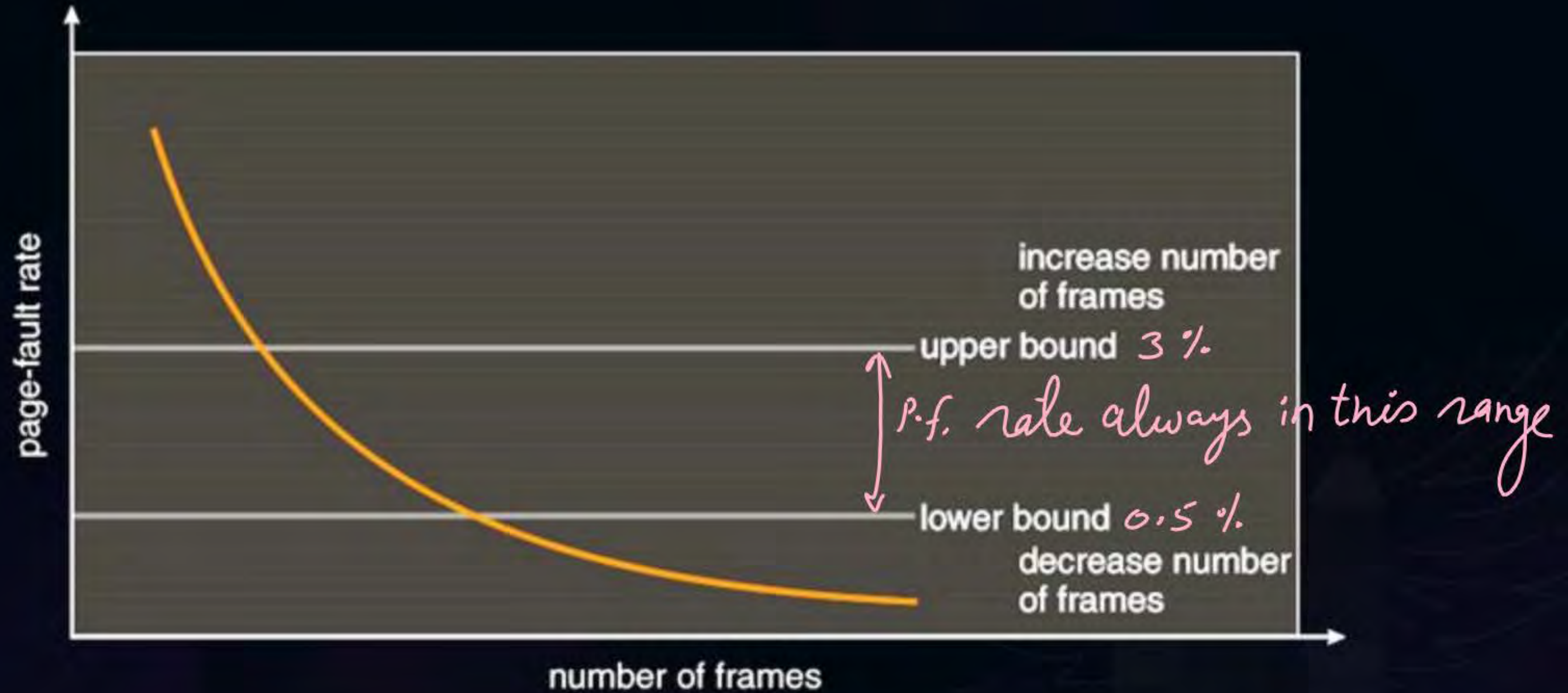


keep working set (current required page) of process in mm.





Topic : Page Fault Frequency





2 mins Summary

Topic

Multilevel Paging

Topic

Frame Allocation



Happy Learning

THANK - YOU

